SECTIONS OF DIRECTED COLIMITS OF SHEAVES VERSUS COLIMITS OF SECTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition definition-separated (canonical tag 007A): a presheaf is separated if restrictions agreeing on a covering come from a common section, i.e. the restriction maps to the sheafification are injective. invoked at proof lines 3688-3690 for the presheaf V \mapsto colim F_i(V) under injective transitions. Makes F' separated so the map to its sheafification is injective.
- Lemma lemma-separated-presheaf-into-sheaf (canonical tag 0082): the canonical map from a separated presheaf into its sheafification is injective. cited at proof lines 3692-3694. Completes the proof of part (1).

Source reference. Original definition 007A: sources/stacks-chow/tags.txt, source lines 274–274. Invoked at proof lines 3688-3690 for the separatedness of the colimit presheaf; load-bearing inside the counted span.

Source reference. Original lemma 0082: sources/stacks-chow/tags.txt, source lines 302–302. Cited at proof lines 3692-3694 to conclude part (1); load-bearing inside the counted span.

Source reference. Original section 002Z: sources/stacks-chow/tags.txt, source lines 116–116. Cited at statement lines 3666-3667 fixing the notion of a system of sheaves over the directed set I; navigation for the setting.

Lemma 0.2. *Let X be a topological space. Let I be a directed set. Let $(\mathcal{F}_i, \varphi_{ii'})$ be a system of sheaves of sets over I, see Categories, Section 002Z. Let $U \subset X$ be an open subset. Consider the canonical map*

$$\Psi : \operatorname{colim}_i \mathcal{F}_i(U) \longrightarrow (\operatorname{colim}_i \mathcal{F}_i)(U)$$

(1) *If all the transition maps are injective then Ψ is injective for any open U.*
(2) *If U is quasi-compact, then Ψ is injective.*
(3) *If U is quasi-compact and all the transition maps are injective then Ψ is an isomorphism.*
(4) *If U has a cofinal system of open coverings $\mathcal{U} : U = \bigcup_{j \in J} U_j$ with J finite and $U_j \cap U_{j'}$ quasi-compact for all $j, j' \in J$, then Ψ is bijective.*

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Proof. Assume all the transition maps are injective. In this case the presheaf $\mathcal{F}' : V \mapsto \operatorname{colim}_i \mathcal{F}_i(V)$ is separated (see Definition 007A). By the discussion above we have $(\mathcal{F}')^\# = \operatorname{colim}_i \mathcal{F}_i$. By Lemma 0082 we see that $\mathcal{F}' \to (\mathcal{F}')^\#$ is injective. This proves (1).

Assume U is quasi-compact. Suppose that $s \in \mathcal{F}_i(U)$ and $s' \in \mathcal{F}_{i'}(U)$ give rise to elements on the left hand side which have the same image under Ψ. Since U is quasi-compact this means there exists a finite open covering $U = \bigcup_{j=1,\ldots,m} U_j$ and for each j an index $i_j \in I$, $i_j \geq i$, $i_j \geq i'$ such that $\varphi_{ii_j}(s)$ and $\varphi_{i'i_j}(s')$ have the same restriction to U_j. Let $i'' \in I$ be $\geq$ than all of the i_j. We conclude that $\varphi_{ii''}(s)$ and $\varphi_{i'i''}(s')$ agree on the opens U_j for all j and hence that $\varphi_{ii''}(s) = \varphi_{i'i''}(s')$. This proves (2).

Assume U is quasi-compact and all transition maps injective. Let s be an element of the target of Ψ. Since U is quasi-compact there exists a finite open covering $U = \bigcup_{j=1,\ldots,m} U_j$, for each j an index $i_j \in I$ and $s_j \in \mathcal{F}_{i_j}(U_j)$ such that $s|_{U_j}$ comes from s_j for all j. Pick $i \in I$ which is $\geq$ than all of the i_j. By (1) the sections $\varphi_{i_j i}(s_j)$ agree over the overlaps $U_j \cap U_{j'}$. Hence they glue to a section $s' \in \mathcal{F}_i(U)$ which maps to s under Ψ. This proves (3).

Assume the hypothesis of (4). In particular we see that U is quasi-compact and hence by (2) we have injectivity of Ψ. Let s be an element of the target of Ψ. By assumption there exists a finite open covering $U = \bigcup_{j=1,\ldots,m} U_j$, with $U_j \cap U_{j'}$ quasi-compact for all $j, j' \in J$ and for each j an index $i_j \in I$ and $s_j \in \mathcal{F}_{i_j}(U_j)$ such that $s|_{U_j}$ is the image of s_j for all j. Since $U_j \cap U_{j'}$ is quasi-compact we can apply (2) and we see that there exists an $i_{jj'} \in I$, $i_{jj'} \geq i_j$, $i_{jj'} \geq i_{j'}$ such that $\varphi_{i_j i_{jj'}}(s_j)$ and $\varphi_{i_{j'} i_{jj'}}(s_{j'})$ agree over $U_j \cap U_{j'}$. Choose an index $i \in I$ which is bigger or equal than all the $i_{jj'}$. Then we see that the sections $\varphi_{i_j i}(s_j)$ of $\mathcal{F}_i$ glue to a section of $\mathcal{F}_i$ over U. This section is mapped to the element s as desired. $\square$